

Part: A

1. Mean, Median, Mode (Customer Income)

I. Mean (Average Income):

- Sum of all customer incomes divided by total customers.

Example: If 5 customers earn [30k, 40k, 50k, 60k, 70k]

$$\text{Mean} = \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

- Where,
- x_i = income of i-th customer,
- n = total customers.

- The mean = $(30+40+50+60+70)/5 = 50k$.

II. Median (Middle Income):

- The middle value when incomes are sorted.
- If n is odd: Median = middle value
- If n is even: Median = average of two middle values

$$\text{Median} = \begin{cases} \frac{x_{\frac{n+1}{2}}}{2}, & \text{if } n \text{ odd} \\ \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}, & \text{if } n \text{ even} \end{cases}$$

- Example: Sorted [30k, 40k, 50k, 60k, 70k] → Median = 50k.

III. Mode (Most Frequent Income):

- Income that appears most often.
Example:
 - If multiple customers earn 40k more than others
 - Mode = 40k.
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2. Standard Deviation vs Variance (Loan Amounts)

- **Variance:** Average of squared differences from the mean. Measures how spread out loan amounts are.

$$\text{Variance} = \sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \text{ (population)}$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \text{ (sample)}$$

1. Compute differences from mean:
 - $10 - 20 = -10$
 - $20 - 20 = 0$
 - $30 - 20 = 10$
2. Square differences:
 - $(-10)^2 = 100$
 - $0^2 = 0$
 - $10^2 = 100$
3. Sum squared differences:

$$100 + 0 + 100 = 200$$
4. Divide by $n = 3$:

$$\sigma^2 = 200/3 \approx 66.67$$

Variance = 66.67k²

- **Standard Deviation:** Square root of variance. Same units as loan amounts, easier to interpret.

$$\text{Std Dev} = \sigma = \sqrt{\text{Variance}}$$

$$\sigma = \sqrt{66.67} \approx 8.16$$

Std Dev $\approx$ 8.16k

3. Random Variable

- **Definition:** A variable that can take different values due to chance.
- **Example:** Default_Status = 1 (Yes), 0 (No)

- $X = \begin{cases} 1, & \text{if defaulted} \\ 0, & \text{if not defaulted} \end{cases}$

- **Probability of X:** $P(X = 1)$ = fraction of defaults
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4. Conditional Probability (Loan Defaults)

Definition: Probability of an event given another event has occurred.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- **Bank Example:**

- Event A = Customer **defaults**
- Event B = Customer has **Credit Score < 600**

$$P(\text{Default} | \text{Credit Score} < 600) = \frac{\text{Number of customers who default AND have Credit Score} < 600}{\text{Number of customers with Credit Score} < 600}$$

- Total customers with Credit Score < 600 → 800
- Number of defaults among them → 320

$$P(\text{Default} | \text{Credit Score} < 600) = \frac{320}{800} = 0.4$$

- Interpretation: 40% of customers with low credit score (<600) defaulted.

5. Bayes Theorem

- **Definition:** Formula to compute conditional probabilities using prior knowledge.

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

- **Bank example:** Probability of default (A) given low credit score (B):

$$P(\text{Default} | \text{LowCredit}) = \frac{P(\text{LowCredit} | \text{Default}) \cdot P(\text{Default})}{P(\text{LowCredit})}$$

6. Empirical Probability vs Theoretical Probability

- **Empirical Probability:** Based on data

$$P(A) = \frac{\text{Number of times event A occurs}}{\text{Total observations}}$$

- **Example:** 600 defaults out of 5000 customers →

$$P(\text{Default}) = \frac{600}{5000} = 0.12$$

- **Theoretical Probability:** Based on assumption/model

$$P(A) = \text{Expected probability from theory}$$

- **Example:** Bank assumes 10% customers default → $P(\text{Default}) = 0.1$

7. Poisson Distribution

- **Definition:** Models number of events in fixed interval (time, space) with known average rate.
- **Business example:** Number of loan applications received per day follows Poisson distribution.

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- **λ:** Average number of events per interval
- **k:** Actual number of events

- **Example:** Number of loan applications received per day.
- If $\lambda = 20$ → probability of exactly 15 applications:

$$P(X = 15) = \frac{20^{15} e^{-20}}{15!}$$

8. Eigenvalues and Eigenvectors (Data Analysis)

- **Definition:**

- Eigenvectors: Directions along which data stretches/compresses.
- Eigenvalues: Magnitude of stretching along that direction.
- For a square matrix A , eigenvector v and eigenvalue λ satisfy:

$$A \cdot v = \lambda \cdot v$$

➤ **Use in Data Analysis:**

- PCA reduces dimensionality
- Eigenvectors → principal directions of data
- Eigenvalues → variance along those directions

Example: Bank may reduce [Income, Loan_Amount, Credit_Score] into fewer components to simplify risk analysis.